DataCo
Supply Chain Review

Executive Overview

Profit $3,966,903 on revenue $33,054,402. 57% of orders arrived late.

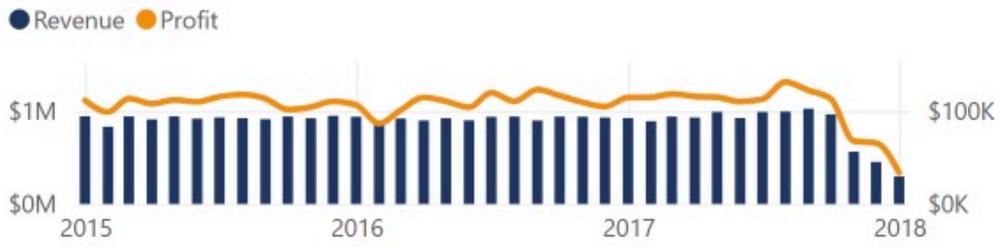
Overview
Delivery
Profitability
Fraud

Year: All
Market: All
Shipping_mode: All

↺ Reset filters

Monthly trend

Revenue | Profit | Orders



Orders by customer segment



Revenue

$33.05M

after $3.73M of discounts

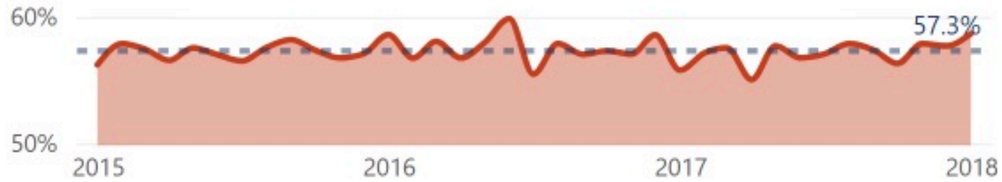
Revenue by market



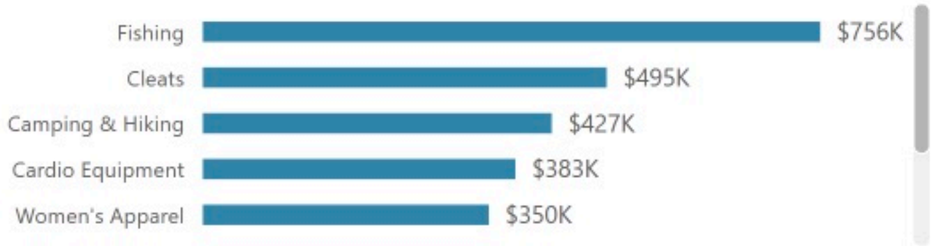
Orders by shipping mode



Late rate by month



Profit by category (top 8)



Profit margin

12.0%

18.7% of items sell at a loss

Suspected fraud, % of profit

18.7%

$741K, all paid by bank transfer

Source: DataCo Smart Supply Chain (Kaggle)
180,519 order items · Jan 2015 – Jan 2018

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Delivery Performance

57% of shipped orders arrived late, by 1.6 days on average.

Late rate

57.3%

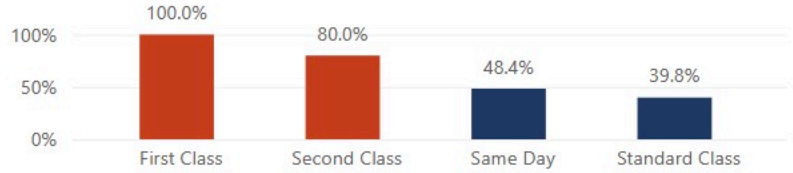
36,048 of 62,897 shipped orders

Late rate after the fix

34.3%

with realistic delivery promises

Late rate by shipping mode



Shipped orders by delivery status

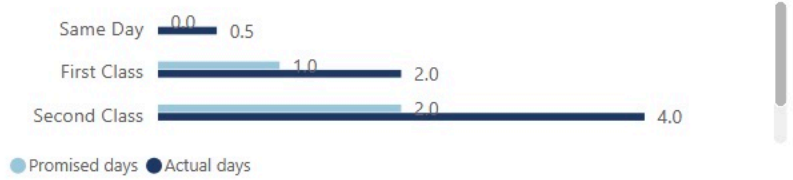


Avg days late

1.6

days late, on average, when lat

Promised vs actual delivery days



Late rate: market × shipping mode

market	First Class	Same Day	Second Class	Standard Class
USCA	100.0%	45.6%	79.9%	40.0%
Pacific Asia	100.0%	47.9%	80.3%	40.4%
LATAM	100.0%	50.8%	79.4%	39.3%
Europe	100.0%	48.2%	80.2%	39.8%
Africa	100.0%	47.1%	80.4%	39.8%

Key finding: lateness is built into the promises, not the regions

- First Class promises 1 day but always takes 2 → 100% late
- Second Class promises 2 days but takes about 4 → 80% late
- Every market shows the same pattern

Recommendation: reset First Class to 2 days and Second Class to 4 days → late rate falls from 57% to about 34%

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Profitability

19% of items sold at a loss. A discount cap would add $1,144,742 profit.

Profit

$3.97M

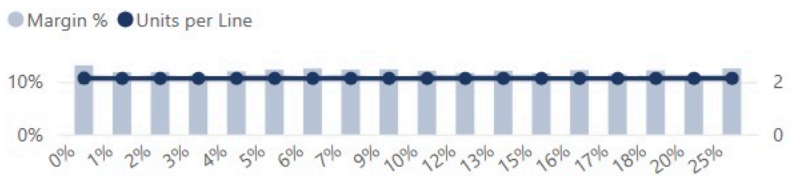
12.0% margin

Losses on loss-making items

-3.88M

on 18.7% of items sold

Bigger discounts don't sell more



What if we capped discounts?

extra profit per year

1.14M

Discount Cap Gain

28.9%

more profit

Top 10 products by revenue

Revenue and Margin % by product_name



Average discount

10.1%

f gross sales given away as discount

Revenue concentration

88.2%

of revenue from the top 9 products

Key finding: discounts give margin away without buying sales

- Units per order line stay at 2.13 at every discount level, from 0% to 25%
- 18.7% of items sell at a loss, costing $3.88M over the period
- The top 9 products bring in 88% of revenue

Recommendation: cap discounts at 10% → about +$367K profit a year (+28% on 2017), then audit costs on loss-making products.

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Fraud Risk

Suspected fraud cost $741,492, which is 19% of profit.

Fraud as % of profit

18.7%
Fraud % of Profit

Suspected fraud value

$741K
1,488 suspected fraud orders

Fraud rate by payment type



Rule test: hold bank-transfer orders until payment clears

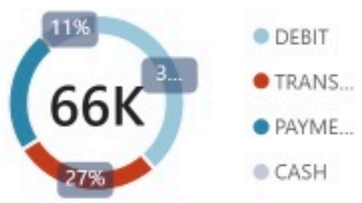
27.5%	100.0%	8.2%	112
orders held	fraud caught	checks that find ...	checks per week

All suspected fraud was paid by bank transfer, so holding transfers catches every case. About 1 in 12 held orders is fraud: roughly 3 checks per working hour.

Fraud value and cases by month



Orders by payment type



Fraud rate

2.3%
of all orders flagged as susp...

Key finding: all suspected fraud comes through bank transfers

- Transfers: 8.2% fraud rate; debit, cash and card payments: 0%
- Fraud costs about $256K a year (2017), a fifth of profit
- No other factor (market, shipping mode, order size) predicts it

Recommendation: hold transfer orders until payment clears → stops all $256K of suspected fraud for about 118 checks a week.

Method note: delivery status was excluded because it is set after the fraud decision (data leakage).

$33.05M

Revenue

12.0%

Margin %

57.3%

Late Rate %

$741K

Fraud Value

Revenue trend

Revenue by Month Start

Category: Fishing

Revenue	Profit	Margin	Items sold at a loss
$6.23M	$0.76M	12.1%	18.5%

Products in this category

product_name	Revenue	Profit	Margin
Field & Stream Sportsman 16 Gun Fire Safe	**$6,226,935.290564688**	**$756,220.7671894992**	**12.**
Total	**$6,226,935.290564688**	**$756,220.7671894992**	**12.**

Revenue and profit by month



Revenue by market

